

LEARNING OUTCOMES!

Candidates should be able to:

- know the distribution of temperature and rainfall, including monsoon, depressions and convectional rain
- know seasonal and regional variations, and the factors contributing to them, including depressions, thunderstorms and cyclones (typhoons)
- understand the causes of the monsoon (knowledge of the causes of other types of rain is not required)
- describe and explain the characteristics of the climate of the arid, semi-arid, humid and highland regions, including seasonal variations
- know the influence of latitude and longitude on day length and climate
- understand the influence of the climate (both the benefits it brings and the problems it causes) on the economy and on the lives of the people: - the influence of low temperature, ice and snow on the lives of people in the mountains - the influence of rain storms and flooding on agriculture, industry and communications - the problems caused by drought and shortage of water supply on agriculture and industry.

WEATHER

- ✓ Weather refers, generally, to day-to-day temperature and precipitation activity.

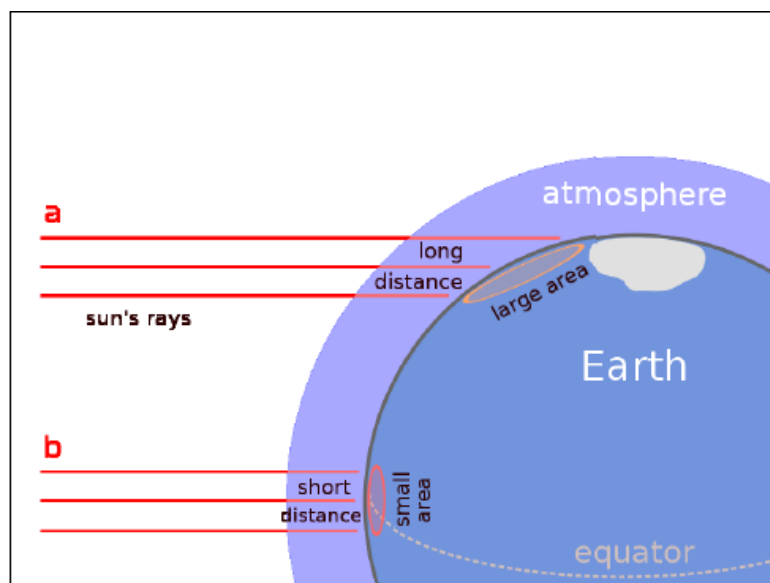
CLIMATE

- ✓ Climate is the term for the average atmospheric conditions over longer periods of time (30 years etc)

FACTORS AFFECTING TEMPERATURE

Angle of Sun

- ✓ Temperature is dependent either directly or indirectly on the influence of the sun.
- ✓ As it is evident the Polar Regions or regions far away from the Equator are generally cooler than the ones nearer the Equator. This is because for solar radiation to reach the poles and heat them it has to travel a larger distance in space, thus it loses its intensity (energy).
- ✓ Also near the poles the radiation arrives at an oblique angle so solar energy spreads over a large area. Since Pakistan lies in subtropical areas it does receive most of intense solar radiation concentrated on a small area thus the temperature in general is warmer. So in general terms Faisalabad will be cooler than say Hyderabad



Influence of large water bodies

- ✓ Water has a high specific heat capacity as compared to land, meaning that water has to absorb more heat to have an increase in temperature as compared to land.
- ✓ Thus land gets hotter quickly than sea etc.
- ✓ Since the land gets hot quickly, the air over it also absorbs heat and gets less dense.
- ✓ This hot air rises and creates a low pressure.
- ✓ The sea on the other hand absorbs less heat and moist air over it gets less hot.
- ✓ Thus it is denser as compared to the air over land and doesn't rise as much as the air over land does, which results in the persistence of high pressure over the sea.
- ✓ Winds travel from high pressure area to low pressure area, thus in this case wind blows from sea towards land.
- ✓ This is known as a sea breeze, which is cool and moist
- ✓ On the contrary at night when there is no solar radiation, the land loses heat quickly and air over it gets cooler and denser.
- ✓ The sea retains its heat and air over it is generally hotter and less dense.
- ✓ So a land breeze occurs when wind blows from land to sea
- ✓ During the summers, the land is much hotter than the Arabian Sea, thus convection currents of air are setup. Cool air from sea flows inland thus regulating the temperature and making summers a bit cooler than it would have been without the influence of the sea
- ✓ In winters, the sea retains its warmth and thus cold air from coastal areas flows towards sea. To replace the air in coastal areas whose rising has created a low pressure, dry warm air from interior Sindh and Punjab flows towards the coast, thus making winters less cold than they would have been without the influence of the sea
- ✓ If an area doesn't experience moderating effect of sea then it has a continental climate, which is characterized by hot summers and cold winters. USUALLY THOUGH NOT ALL THE TIMES, most of the rainfall in these continental climate areas occurs in the winters

Altitudes

- ✓ Altitude also has a big impact on the average temperatures experienced by an area.
- ✓ The solar radiation is a short-wavelength radiation, which does not heat the atmosphere directly when it passes through it.
- ✓ This is because it is poorly absorbed by the air particles (gases and water vapours).
- ✓ Instead the atmosphere is heated from the bottom by long-wave radiation from the earth's surface.
- ✓ The Earth's surface reflects the short wavelength radiation as long wavelength radiation.
- ✓ This long wave radiation is better absorbed by gas molecules and water vapours as compared to the short wavelength radiation.
- ✓ The temperature of the atmosphere decreases with increase in altitude, as short wavelength has to pass its heat to a lot of air particles to transfer heat to the high altitude areas.
- ✓ Also as the altitude increases the amount of atmosphere decreases (air gets thinner and thinner), thus there are less gas molecules and water vapours present to absorb the long wavelength radiation.
- ✓ Thus overall a smaller proportion of the total heat energy is passed on to the higher parts of the atmosphere
- ✓ On average dry air loses around 9.8°C per every Km rise in height, meanwhile moist air loses around 10°C for every Km rise in height

Valley

- ✓ Most of the sun's rays are blocked from reaching the lower parts of valleys due to the presence of mountains, which surround the valley.
- ✓ This lowers the average temperatures of these valleys, as less solar radiation reaches the surface of the valley to be reflected and then absorbed.

Cloud Cover

- ✓ Temperatures decrease when clouds appear and block out the Sun's rays, which are then reflected back into space.

- ✓ However cloud formation only takes place when air contains sufficient moisture and has been sufficiently cooled.
- ✓ Thus cloudy days are much cooler than sunny days during the same months.
- ✓ However cloudy nights are hotter than cloudless nights because the clouds trap heat inside the Earth's surface. This heat is easily radiated back to space when no clouds exist but when they do exist this heat is reflected back onto the Earth's surface.
- ✓ These two reasons help us explain why a day in desert is very hot 50°C but night is cool/cold 5°C

MONSOON RAINFALL

- ✓ Monsoons are seasonal winds that blow from sea towards land from July to August (summers).
- ✓ After it they reverse their direction and blow from land towards the sea (winters).
- ✓ During the summers due to the high angle of sun, Punjab and Sindh plains get heated up, thus an area of very low pressure develops here as the hot air rises.
- ✓ There are two monsoon systems, one originating from the Bay of Bengal and other from the Arabian Sea.
- ✓ During summer low air pressure is developed over Pakistan.
- ✓ While high air pressure prevails over sea (Bay of Bengal, Indian Ocean and Arabian Sea) so moist (humid/wet) wind from Indian Ocean and Arabian Sea blow towards land to fulfill the low air pressure.
- ✓ Wind from Indian Ocean reaches Pakistan after crossing India called primary monsoon.
- ✓ Monsoon winds from Arabian Sea are called secondary Monsoon which causes less rain.
- ✓ These winds cause rain from July to September.
- ✓ The amount of rain decreases from northern area to Southern areas.
- ✓ Monsoon of Bay of Bengal affect northern Punjab, Khyber-Pakhtunkhwa, Gilgit-Baltistan and Azad Kashmir
- ✓ On the other hand another monsoon system that originates in Arabian Sea also travels inland but doesn't bring significant rainfall to areas of Sindh because of a temperature inversion layer

Describe the causes of the south west monsoon.

- ✓ Coming from Arabian Sea
- ✓ Sun heats up in (tropical) continents (land) faster or more quickly than the surrounding oceans (water);
- ✓ Warm air rises;
- ✓ Attracts cool, moist air from the sea;
- ✓ Rain bearing winds push further inland causing (heavy) rain

Explain two impacts of a heavy monsoon. You should develop your answer.

- ✓ Floods;
- ✓ Heavy rain; causes poor visibility and accidents (dev);
- ✓ Roads become rivers; cannot travel to work or school / towns and cities cut off;
- ✓ Flights cancelled; negative impact on trade / tourism / business;
- ✓ Crops destroyed; which causes food shortages / can lead to famine;
- ✓ Water level in reservoirs or dams rises; leading to more water for irrigation / domestic / industrial use / no water shortages / HEP;
- ✓ Businesses / markets are closed; leading to loss of income / jobs disrupted;
- ✓ Homes washed away / flooded / buildings destroyed;
- ✓ People are injured / killed; pressure on healthcare services
- ✓ Stagnant water; causes diseases;
- ✓ Waterlogging; impacts on the economy as Pakistan is an agriculturally based country;
- ✓ Replenishes groundwater; reduces water shortages;

WESTERN DEPRESSION

- ✓ These are wind systems that originate from the Mediterranean Sea and travel eastwards towards Afghanistan and Pakistan.
- ✓ Since they make a long journey they lose most of their water when they reach Pakistan.
- ✓ During the winters (Late November to March) they bring rainfall because during winters the Arabian Sea retains its warmth and thus cold air from coastal areas flows towards sea.
- ✓ The flow of air from coastal areas creates a low pressure, thus dry colder air from interior Sindh and Punjab flows towards the coast.
- ✓ To replace the low pressure formed in Punjab and Sindh plains, these western depressions move from high pressure area (Mediterranean Sea) towards the low pressure area (Indus Plains)

CONVECTIONAL RAINFALL

- ✓ Convection rainfall occurs when the sun heats up the land.
- ✓ The air near the land absorbs heat by conduction, and thus gets less dense and rises.
- ✓ The moisture in air condenses to form clouds, and when this air is cooled to a certain minimum level rainfall occurs

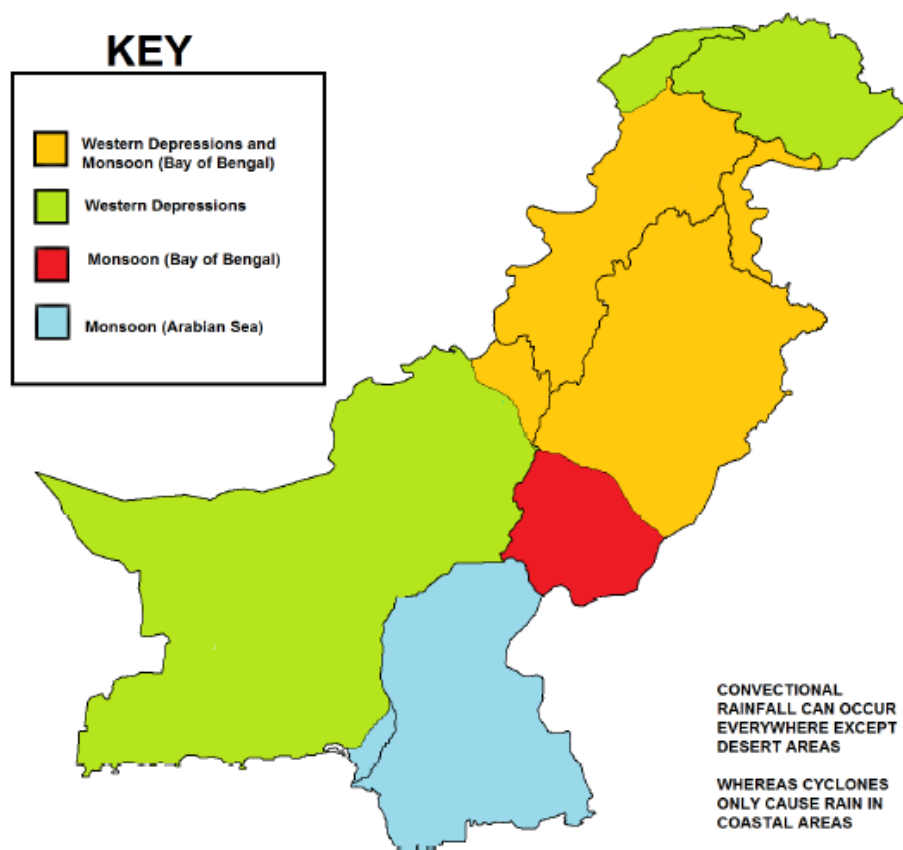
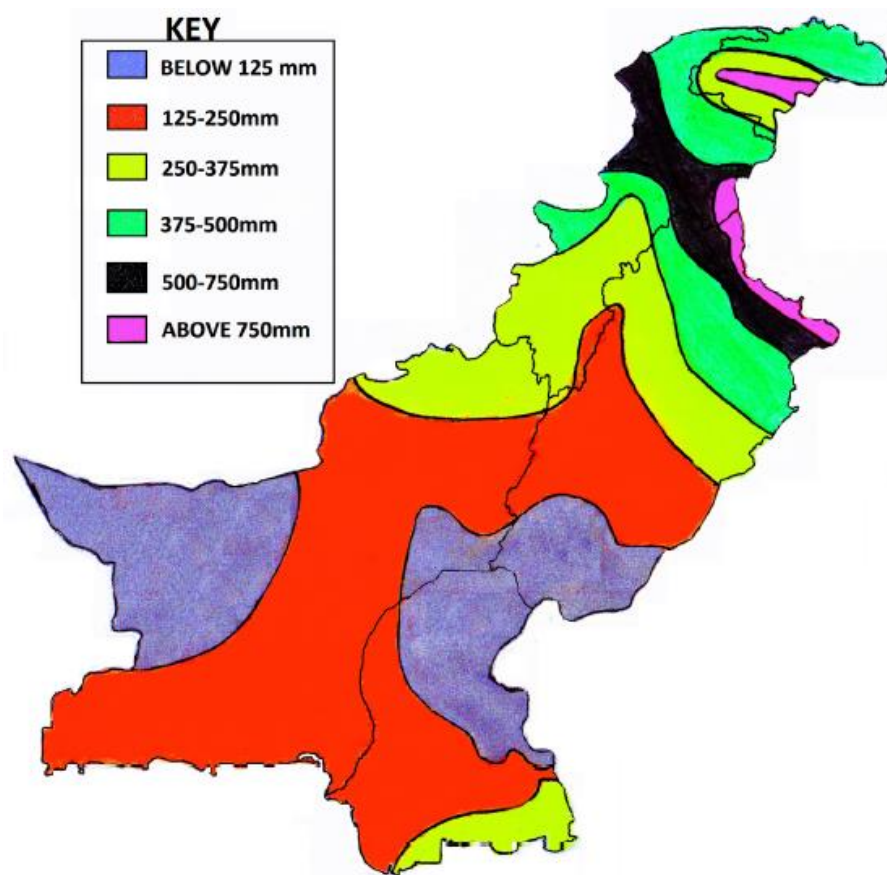
THUNDERSTORM

- ✓ Thunderstorms occur in the same fashion as convectional rainfall with the main difference being that the air here rises very quickly forming Cumulonimbus clouds, which are very tall and big.
- ✓ This is followed by strong winds, lightning and heavy rainfall.
- ✓ Hailstones may also be formed, when the water vapours are cooled multiple times (to form water first and then ice).

CYCLONE

- ✓ Cyclones are formed when the average temperature of large water body exceeds 27° C, thus forming an area of very low pressure.
- ✓ This area then draws further air from high pressure areas quickly.
- ✓ A cone is formed when two oppositely travelling cold and warm winds meet.

- ✓ Intense lightning followed by very strong winds and heavy rainfall occurs.



FLOODS

- ✓ Flood is a temporary covering of land by water which is not usually under water.
- ✓ They can be caused by variety of factors like Heavy monsoon rainfall accompanied with melting of snow and ice in glaciers, dam or barrage failure, unusually high tides in coastal areas and by a tsunami
- ✓ However their effects can be exacerbated by factors like deforestation, failure to heighten or strengthen embankments, poor medical and communication facilities in aftermath of disaster and inadequate warning systems to allow people time for escape.
- ✓ Benefits of floods are limited to restoring underground water supplies, filling reservoirs of dam and laying down a fresh layer of silt on the inundated soil; which makes the soil fertile

Effects

- ✓ Casualties - People and livestock die due to drowning. It can also lead to epidemics and waterborne diseases.
- ✓ Suspension of Trade- Airports can be closed (businessmen can be affected), roads and railway lines severed causing disruption to supplies of raw materials and products etc
- ✓ Water supplies can be contaminated.
- ✓ Spread of water-borne diseases. This could pose a serious danger to health of people who are affected
- ✓ Crops can be washed away, thus leading to food shortages.
- ✓ There could be a shortage of supplies to agro-based industries, like shortage of raw cotton to cotton factories. These factories thus won't be able to fulfil their orders and will lose customers. To reduce their operating costs they will lay off workers causing unemployment and their own profits will decrease
- ✓ Silt deposited by floods adds nutrients to the flooded soil
- ✓ Trees die as their submerged roots can't absorb oxygen
- ✓ There could be a decline in tourism, rebuilding costs could soar etc

Solution

- ✓ Dams could be built to contain and regulate the flow of water and prevent flash floods
- ✓ Advanced warning systems should be installed in flood prone areas to warn people to get out before it's too late

- ✓ Medical and transport facilities be regularly updated and checked
- ✓ Embankments and levees should be heightened and strengthened
- ✓ Also afforestation and re-afforestation projects should be carried out in Northern Mountains to reduce run-off and thus reducing chances of flash floods
- ✓ Mangrove plantations should be protected and allowed to grow, which protect from damaging rise in tide level during a tsunami

DROUGHTS

- ✓ A drought is a period extending over several months or years when a region suffers a deficiency in its water supply.
- ✓ This occurs when a region receives below than average precipitation over a period of time.
- ✓ Meteorological drought is brought about when there is a prolonged period with less than average precipitation.
- ✓ Agricultural droughts are droughts that affect crop production. This condition can also arise independently from any change in precipitation levels when soil conditions and erosion triggered by poorly planned agricultural endeavours. These cause a shortfall in water available to the crops.
- ✓ Hydrological drought is brought about when the water reserves available in sources such as aquifers, lakes and reservoirs fall below the statistical average.

Effects

- ✓ Crop yields can be dramatically reduced and livestock production could fall (milk, eat etc). This can raise prices of goods and affect price of exported goods making them uncompetitive in international markets thus harming national interests
- ✓ Dust storms can occur, when drought hits an area suffering from desertification and erosion. This can lead to siltation in reservoirs of dam, hampering electricity generation and weakening dam's foundation
- ✓ Habitat damage, affecting both terrestrial and aquatic wildlife
- ✓ Malnutrition, dehydration and related diseases could affect millions
- ✓ Mass migration causing the increase in internal refugees or international refugees
- ✓ Reduced electricity production could occur as reduced water flow through hydroelectric dams leads to low industrial production and less exports resulting in less foreign exchange
- ✓ Shortages of water for industries like (juice etc), which affects employment and GDP
- ✓ Social unrest may follow leading to instability, which can discourage foreign investment thus local sectors may suffer from outdated machinery techniques etc resulting in low production
- ✓ War could occur over natural resources, including lakes and fertile areas etc
- ✓ Wildfires can become common and can cause health hazard to people

